

JiaYu Zhao

Electronic Information Engineering | FPGA, Embedded Systems
and Communications

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EDUCATION

UNIVERSITY

Harbin University of Science and
Technology

DEGREE

B.Eng. in Electronic Information
Engineering

DATES

2021 - 2025

COURSEWORK

Signals and Systems, Digital Circuits,
Analog Electronics, C Programming,
Microcontrollers

AWARDS AND CREDENTIALS

University Scholarship
Awarded during undergraduate study.

Second Prize in Mathematical Modeling
Accurate localization of multiple rocket
debris objects.

CET-4

Technical reading and English
knowledge-base practice.

LEADERSHIP

Youth Volunteer Association | Team
Captain

2022.01 - 2023.12

Coordinated event support, marathon
operations and elder-care volunteering;
developed task ownership and
teamwork.

TECHNICAL STACK

FPGA: Verilog, Quartus, ModelSim,
TimeQuest, DDS, QPSK, UART, I2C

Embedded: C, STM32 HAL, FreeRTOS,
PWM, ADC, DMA, encoders, PID, Flash

DSP: MATLAB, BER, FFT, sampling,
spectra, filtering, features

Software: Python, Git, Linux, Docker,
SQLite/MySQL, CMake, documentation

PUBLIC CONTACT

No phone number, age, gender, salary
expectation, preferred city or previous
email is included in this public version.

PROFILE

Electronic information engineering graduate building practical workflows across communications, DSP, FPGA RTL, STM32/FreeRTOS and Python. Projects retain source, tests, constraints, build evidence, scripts, documentation and known limits.

- Break systems into independently verifiable algorithm, timing, interface and presentation layers.
- Debug from waveforms, logs, serial data, scripts and build reports.
- Separate simulation, compilation, timing closure and physical verification claims.

SELECTED PROJECTS

FPGA QPSK Baseband Link and Visualization

Verilog | UART | Python | Cyclone IV E

PRBS7, Gray mapping, discrete phase channel, hard decisions, BER counting and UART/CSV telemetry with Python constellation and I/Q views.

- Resolved pipeline alignment, signed widths and UART throughput limits.
- ModelSim and CSV paths pass; Quartus timing checks produce a .sof image.

[Compilation and simulation complete](#); [physical UART confirmation pending](#).

Multi-Function FPGA DDS Signal and Measurement System

DDS | I2C | PCF8591 AD/DA | 74HC595

Four-waveform DDS, ADC/DAC access, sample statistics and a six-digit display in one synthesizable Cyclone IV E design.

- Implemented a 32-bit phase accumulator, 256-entry LUT, I2C state machine and exponential average.
- ModelSim passes; Quartus synthesis, fitting, assembly and TimeQuest complete.

[Compilation and simulation complete](#); [analog board measurement pending](#).

STM32F103 FreeRTOS Multi-Mode Smart Car

C | FreeRTOS | PID | Encoders | C8T6/RCT6

Dual-wheel PID, odometry, line following, avoidance, Flash parameters, safety protection and Bluetooth command firmware.

- Seven task areas around a 10 ms control loop with diagnostics and watchdog protection.
- Both targets cross-build; PID and protocol host tests pass; ELF/HEX/BIN/MAP artifacts produced.

[Software build complete](#); [vehicle electrical, mechanical and calibration tests pending](#).

Mathematical Modeling: Accurate Localization of Multiple Rocket Debris Objects

[Mathematical Modeling](#) | [Localization Analysis](#) | [Second Prize](#)

Modeled and analyzed the accurate localization of multiple rocket debris objects.

- Translated the real-world positioning problem into solvable constraints and completed the competition submission.

[Competition project](#); [Second Prize in Mathematical Modeling](#).

ENGINEERING METHOD

Define acceptance criteria first, verify modules independently, retain evidence at each boundary, and package source, constraints, scripts, results and limitations for reproducible handoff.